INSTITUTE OF INFORMATION TECHNOLOGY · UNIVERSITY OF DHAKA

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Terra-Audit

An AI-Assisted Digital MRV Platform for AWD Rice Irrigation and Agricultural Land Management Carbon Credits

Digital MRV · AWD rice · Sentinel-1 SAR · Carbon accounting · Evidence provenance · Readiness · Internal review

PRESENTED BY
Kazi Nahid
BSSE Roll 1437 · IIT, University of Dhaka

SUPERVISED BY
Dr. Emon Kumar Dey
Professor, IIT, University of Dhaka

METHODOLOGIES
Verra VM0051 · VM0042
+ VMD0054 v1.1 leakage adapter (scoped)

DEPLOYED
terra-audit.vercel.app
Next.js · FastAPI · PostgreSQL · worker

1 Introduction & Background

Flooded rice paddies are a major source of agricultural **methane**: waterlogged soil turns anaerobic and favours methane-producing microbes. **Alternate Wetting and Drying (AWD)** lets the water drop below the soil surface before re-irrigating, cutting methane while maintaining yield. Free **Sentinel-1 radar** sees through cloud and shows when a field is flooded.

1.00 → 0.55 Daily CH_4 scaling factor, continuous flooding → true AWD (≥ 2 drydowns): **45% lower**. IPCC 2019 Table 5.12 · VM0051

2 Problem Statement

Practice claims are hard to verify
Irrigation logs are incomplete; sensors on every smallholder plot are too costly.

Evidence gets separated from the number
Spreadsheets lose the factors, methodology version and history behind a result.

Verification needs traceability
Verifiers must trace every tonne of CO_2e back to its evidence and rule.

Gap: remote-sensing tools and carbon spreadsheets live apart; reviewers reconstruct by hand which plots, logs and methodology version produced a figure.

3 Research Objectives

Detect AWD from space
Flooding & drydown from free Sentinel-1 SAR, labelled as estimates.

Scoped carbon engines
VM0051 v1.1 (rice, QA3) and VM0042 v2.2 (cropland ALM) with visible intermediates.

Cropland evidence
Practices, production, soil sampling, lab, custody & livestock records.

Readiness & leakage
Missing or unsupported rules shown, never silently counted as met or zero.

Frozen, versioned results
Every calculation freezes its evidence, inputs and results.

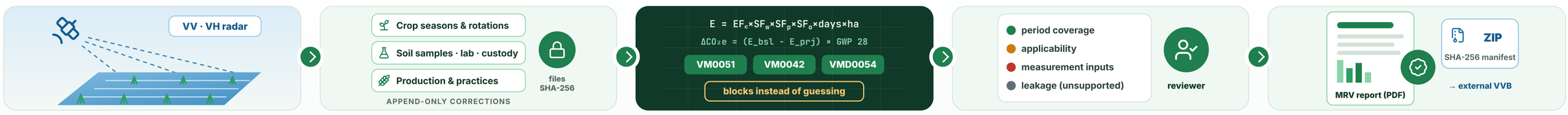
Internal review
A second person approves before a result reaches a report.

AI as an aid only
ML detectors & LLM explanations never change a calculated value.

Multi-user platform
Org & project access control, background jobs, history & export.

4 Graphical Abstract

FROM SATELLITE EVIDENCE TO A VERIFIER-READY CARBON REPORT



01 · MEASURE FROM SPACE
Sentinel-1 radar
Earth Engine time series show when each field was flooded and drained, without field sensors.

02 · KEEP EVIDENCE TOGETHER
One evidence record
Seasons, soil, production and files stay linked to the field; corrections are append-only.

03 · CALCULATE BY THE RULES
Deterministic engines
Verra equations with cited IPCC factors; every intermediate value is kept.

04 · CHECK, THEN REVIEW
Readiness & review
Checks block incomplete work; a second person approves or requests changes.

05 · REPORT FOR THE VERIFIER
Verifier-ready package
MRV report (PDF) plus an evidence ZIP with a SHA-256 manifest.

2 drydowns → 2.12 tCO_2e / ha Live run on the deployed app: Sentinel-1 detected 2 AWD drydowns in an 84-day season. The VM0051 engine returns a **net** reduction after the 15% uncertainty and N_2O deductions.

Advisory AI lane: explains, never decides
Random-forest cross-check and cited LLM explanations go to a human reviewer. No AI component can change a value, mark a requirement compliant, or create credits.

5 Materials & Methodology: End-to-End Workflow

SIX STEPS, EVERY ONE SAVED WITH ITS EVIDENCE

01 **Register field** — Draw, upload (GeoJSON/KML) or paste the boundary; area, district and type are auto-detected. FIELD TEAM

02 **Record season** — Crop, dates, rotations, fallow or missing periods and attachments; corrections are appended. FIELD TEAM

03 **Satellite analysis** — A queued job pulls Sentinel-1, smooths it (Savitzky–Golay) and counts drydowns. AUTOMATIC · WORKER

04 **Calculate** — The engine runs on a frozen snapshot; issuance gates and readiness are evaluated. ONE CLICK

05 **Review** — A reviewer (never the submitter) records findings, diffs versions and decides. REVIEWER

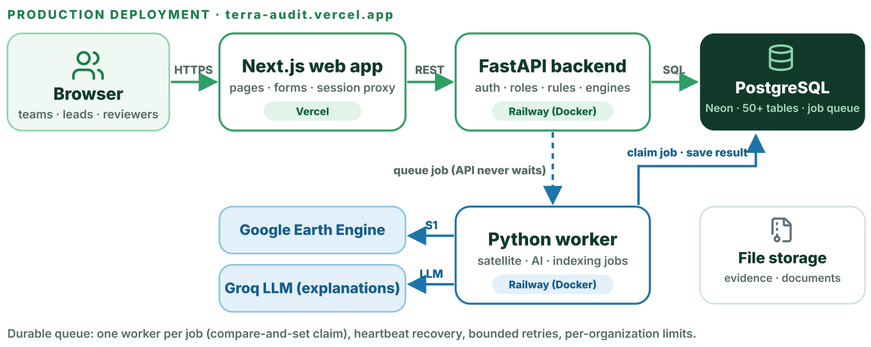
06 **MRV report** — Project PDF and a SHA-256-hashed evidence package for an external VVB. PROJECT LEAD

FROZEN AT COMMIT (IMMUTABLE SNAPSHOT)
Field boundary & area · Crop seasons & engine inputs · Methodology bundle & factors · Readiness result · Evidence & provenance

CALCULATION & REVIEW STATES
`draft` → `ready_for_review` → `superseded`
`submitted` → `in_review` → `internally_approved`

READINESS STATES PER REQUIREMENT
satisfied · needs review · missing · unsupported (stays visible) · not applicable

6 System Architecture & Experimental Setup



AI provider adapters: **fake** (tests), **self-hosted**, **Groq** and **OpenAI**, behind one interface. Methodology PDFs are indexed by section and equation for retrieval.

7 AI Authority Boundary

CODE DECIDES · AI EXPLAINS

AUTHORITATIVE · DETERMINISTIC
Project & evidence → Data validation → VM0051 / VM0042 engine → Results & readiness

ADVISORY · AI
Satellite, ML & crop models → Bundle-scoped context packet → LLM provider → Output validator → Draft explanation → Human reviewer

Bounded context
The model sees only the active methodology bundle, its corrections, the evidence and the calculation.

Validated output
14 named checks: cited sentences must exist, numbers must match, "certified" or "approved" claims are rejected.

Fail closed
One repair attempt, then fail; nothing partial is ever saved.

No AI component can change a calculated value, mark a requirement compliant, or create registry credits.

8 Results & Analysis

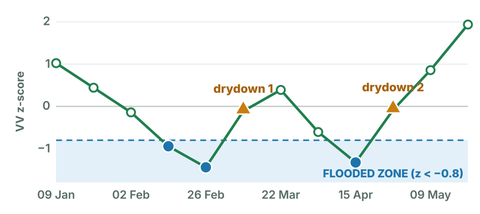
ALL VALUES FROM THE LIVE APP AND THE TEST SUITE

A Water regime drives the reduction



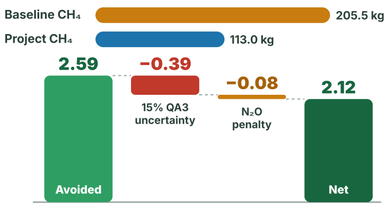
Water-regime factor SF_W, IPCC 2019 Table 5.12 (VM0051): 2 drydowns cut daily CH_4 by 45%.

B Sentinel-1 drydown detection (live run)



● flooded ▲ drydown (> 1.2σ rise after a flooded date) · 12 scenes, Jan–May 2026 → **2 AWD events · 84-day season**

C Worked VM0051 calculation



tCO_2e · 1 ha · 84 days · 2 drydowns · 100 kg N/ha · avoided 92.5 kg CH_4 × GWP 28 · EF_c 0.85 kg CH_4/ha/day.

D ML cross-check of the detector

863 labelled plots → 688 train / 172 held-out

Random forest · Sentinel-1 + Satellite Embedding

72.7%	71.1%	0.719	0.754
PRECISION	RECALL	F1	ROC-AUC

AWD plots correctly identified (held-out): 64 / 90

Repeated 5×5 cross-validation: **66.3% ± 4.2%**. Data: microsoft/rice-irrigation-mapping-s1s2 (Punjab labels, not Bangladesh). Advisory only: it flags disagreement and never changes a calculation.

E Readiness gate on a VM0051 field



1 satisfied · 3 needs review · 1 missing · 3 unsupported

- `monitoring_period_coverage`: crop seasons cover the period
- `methodology_applicability`: crop outside usual scope, reviewer confirms
- `evidence_review_status`: observations not yet accepted by a second person
- `required_measurement_inputs`: AWD events & season length not supplied
- `leakage_assessment`: not implemented, stays blocking
- `n2o_baseline_fertilizer`: not implemented, shown to the verifier

Example run on the deployed app (6 of 8 checks listed). Unsupported rules are never silently satisfied; while anything is missing or needs review the version stays `draft`.

F Software verification

184 backend tests · 0 failed

- 4 / 4 frontend tests
- Production build
- TypeScript check
- 0 lint errors

ID	SCENARIO → EXPECTED	
TC-01	500,000 ha field → rejected above 60,000 tCO_2e/yr	PASS
TC-03	Another organization reads a field → 404	PASS
TC-04	PDF after evidence edits → built from frozen snapshot	PASS
TC-06	AI claims "certified" → rejected, nothing saved	PASS

pytest, npm test, next build, tsc and ESLint re-run on 10 Oct 2026.

Not claimed (not executed): Independent AWD field ground truth · Browser end-to-end tests · Live PostgreSQL migration verification · 20-case AI evaluation

Software verification ≠ VVB verification ≠ Verra issuance

9 Discussion

Free radar can surface AWD. Sentinel-1 drydown signatures were detected without field sensors, but they remain satellite estimates: a field can drain and re-flood between two passes.

Reproducibility comes from freezing. Calculating only from an immutable snapshot means a report rebuilt later shows the same numbers and evidence (TC-04).

Failing closed beats guessing. Gates block oversized projects, unsupported rules and incomplete leakage instead of reporting zero. Golden-value tests caught a wrong emission factor (1.4 → 0.85).

ML is a second opinion, not proof. About 71% of unseen plots are classified correctly with out-of-region labels, so the model stays advisory.

10 Conclusion

Terra-Audit shows that the evidence behind an agricultural carbon claim can be **collected, calculated under VM0051 and VM0042, and reviewed in one system**, with every result traceable from field evidence to the final estimate and the calculation kept separate from the AI around it. A result the implemented methodology cannot support is blocked, not reported.

Satellite AWD evidence · 2 Verra engines + leakage · Immutable versions · Two-person review · Hashed MRV package · Bounded, validated AI

11 Future Scope

P0
- Fix CI ordering and close endpoint authorization gaps
- Resolve the additionality/readiness contradiction without weakening the gate

P1
- **Field ground-truth campaign** for AWD in Bangladesh / South Asia
- Per-calculation rice provenance and a tamper-evident evidence manifest
- Complete the 20-case AI evaluation; verify PostgreSQL, storage and worker recovery

P2
- Versioned migrations; Eq. 38 uncertainty for large projects
- Remaining VM0051 terms, extended only with expert or VVB review

12 · KEY REFERENCES

[1] Verra, *VM0051 v1.1: Improved Management in Rice Production Systems.*
[2] Verra, *VM0042 v2.2: Improved Agricultural Land Management* (incl. 11 Jun 2026 corrections).
[3] Verra, *VMD0054 v1.1: Leakage from the Displacement of Agricultural Activities.*
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[5] IPCC, *AR5 Climate Change 2013: The Physical Science Basis*, GWP100 values.
[6] ESA / Copernicus, *Sentinel-1 SAR User Guide*; Google Earth Engine (COPERNICUS/S1_GRD).
[7] Lampayan et al., "Adoption and economics of AWD …," *Field Crops Research* 170, 2015.
[8] Savitzky & Golay, "Smoothing and differentiation of data …," *Anal. Chem.* 36(8), 1964.
[9] Breiman, "Random Forests," *Machine Learning* 45, 2001; Chen & Guestrin, "XGBoost," KDD 2016.
[10] Microsoft, *rice-irrigation-mapping-s1s2* dataset (Punjab AWD labels).

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Next.js · FastAPI · PostgreSQL · Python worker · Google Earth Engine · scikit-learn / XGBoost

Live app
terra-audit.vercel.app

Source code
github.com/knah1d/terra-audit